

STATEWISE BUSINESS ORDERS TRENDS ANALYSIS – INDIA MARKET

1. PROJECT OVERVIEW:

This analysis focuses on understanding order trends across Indian states, product categories, and sub-categories by consolidating key performance indicators into a unified dashboard. It provides a holistic view of sales, profitability, and target achievement, enabling stakeholders to evaluate performance and identify actionable insights.

2. OBJECTIVES:

These are the objective were going to found from this analysis

- Evaluate how each product category (Clothing, Furniture, Electronics) performs against defined sales targets to identify strengths and gaps in achievement.
 - Analyse the correlation between sales volume and profitability to determine optimal selling quantities and detect loss-making patterns.
 - Compare order distribution across states to highlight high-demand regions and uncover underperforming markets for strategic focus.
 - Track sales performance over time to identify seasonal fluctuations, declining trends, and opportunities for corrective action.
 - Assess contribution of individual sub-categories (e.g., Printers, Sarees, Phones) to overall revenue, enabling product-level prioritization.
 - Visualize geographic sales distribution across cities to support regional marketing strategies and resource allocation.
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3. DATA SOURCES:

Time line:2018-2019.

Domain: E commerce data analysis

4. DATA CLEANING (PREPROCESSING):

The following preprocessing steps were performed to ensure data quality:

- Handled missing values
- Removed duplicate records
- Corrected and changed data types
- Renamed columns for better readability
- Delimit the columns
- Sort and filter the column
- Ensured uniqueness across key identifiers for reliable aggregation.

TOOLS USED:

- Excel
 - Power BI (power query)
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5. DATA TRANSFORMATION:

To enhance analysis, the following transformations were applied:

- Created a dedicated Calendar Table to enable robust time-based analysis
- Developed custom columns to enhance analytical depth and improve reporting accuracy.
- Derived key performance indicators (KPIs).

DAX Measures:

These are the key measures identified in this analysis

- ❖ **Total Orders** → 1.5K
 - ❖ **YTD Sales** → 159K
 - ❖ **Average Profit** → 24K
 - ❖ **Total Sales Amount** → 432K
 - ❖ **Total Target Amount** → 436K
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6. DATA MODELLING AND DAX (POWER BI):

Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.

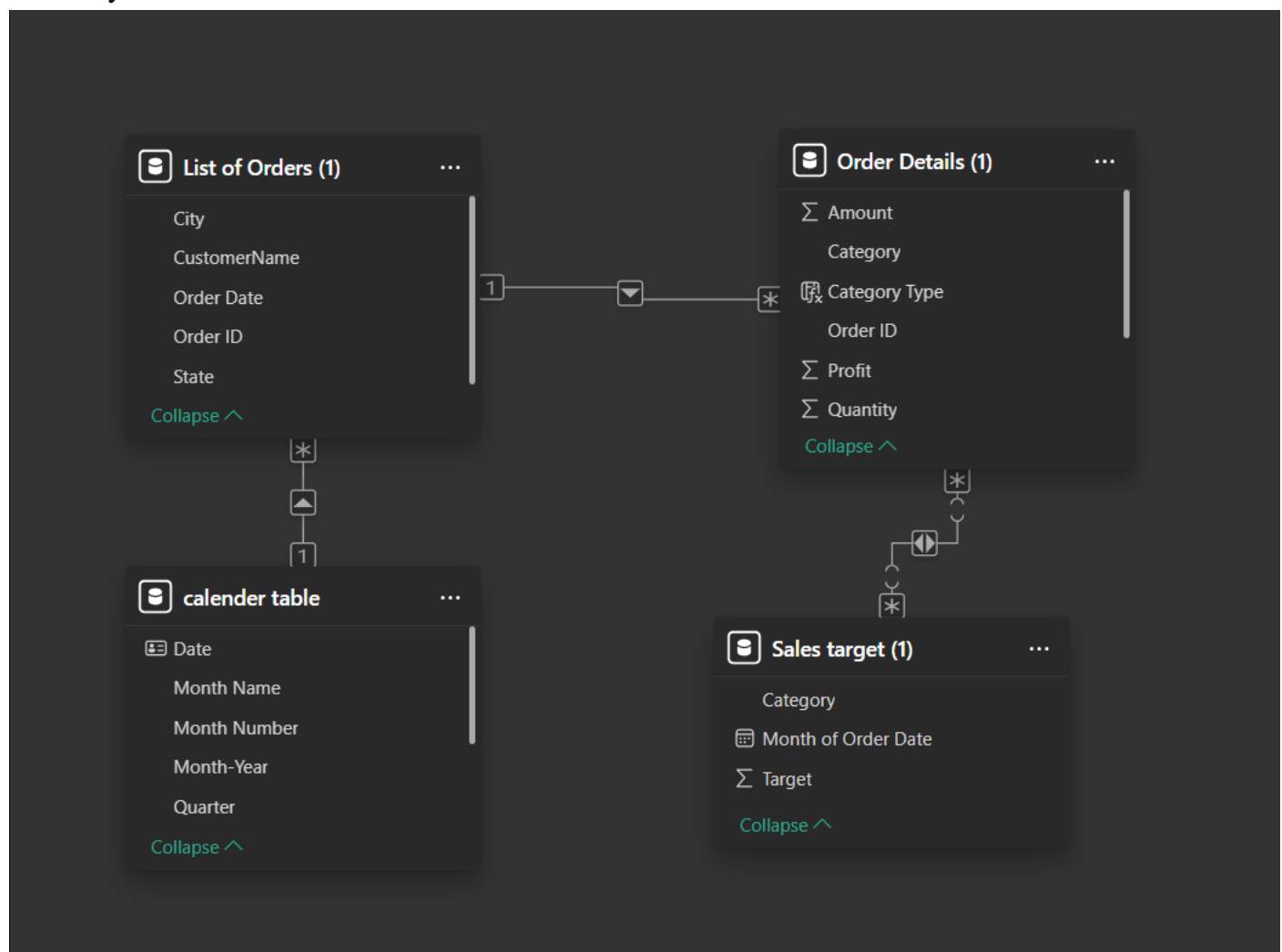


FIG: data modelling

7. EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION:

- Performed statistical analysis
- Built interactive dashboards and visualizations

Features Included:

- Filters (Slicers)
- Interactive visuals

Multiple Visualizations based on problem statement

- Sales Target Achievement by Category → *Bar Chart*
- Profit vs Quantity Relationship → *Scatter Plot*
- Orders by States → *Horizontal Bar Chart*
- Monthly Sales Trend → *Line Chart*
- Sales Distribution by Sub-Category → *Tree map*
- Total Amount by Sales (Geographic View) → *Map of India*

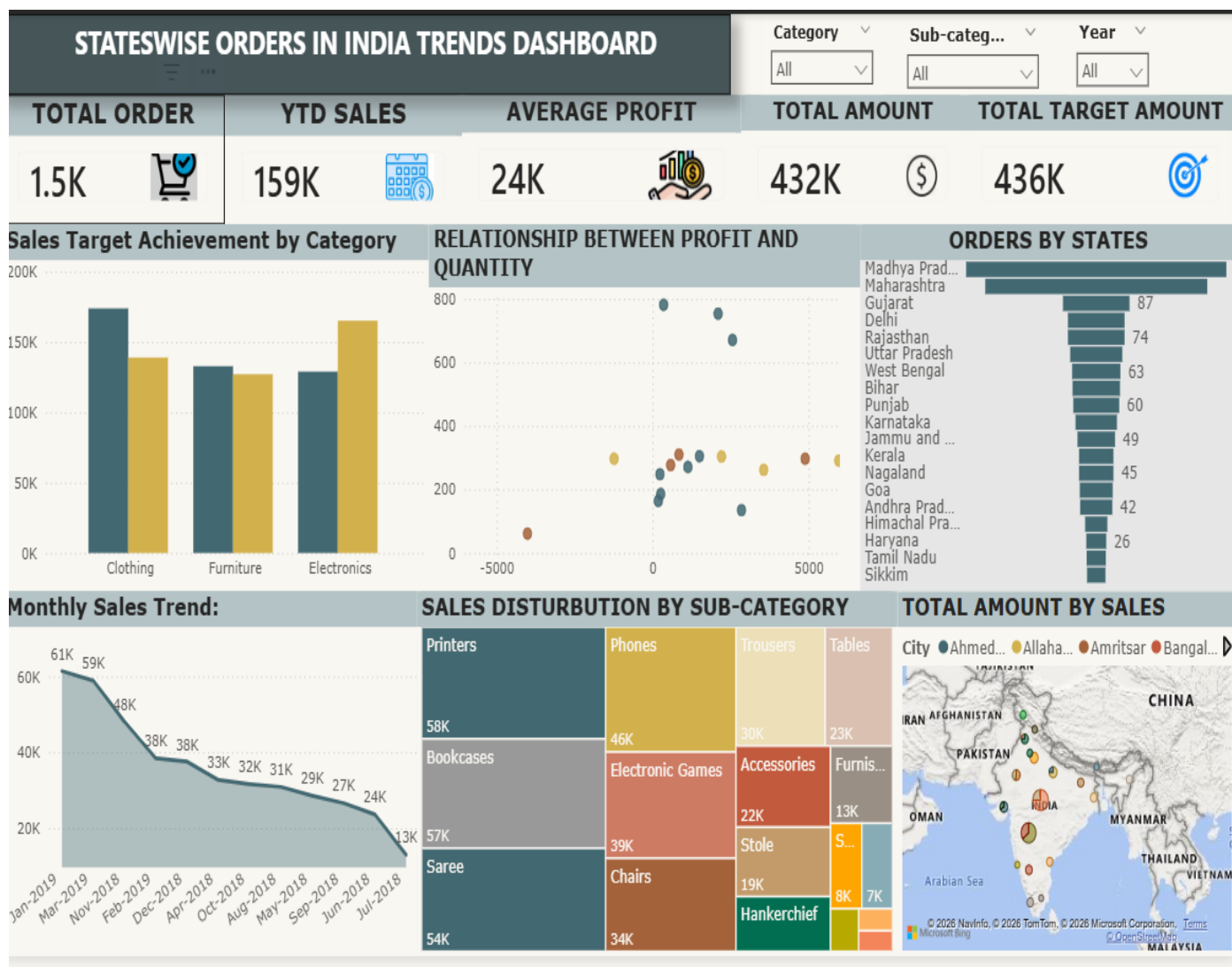


FIG: FINAL DASHBOARD

8. KEY INSIGHTS (SMART + ANALYTICS APPROACH):

The following KPIs were identified through the analysis.

8.1. DESCRIPTIVE ANALYTICS:

- **Clothing and Furniture** nearly met targets, while **Electronics** fell short by **4K**. Specific campaigns can close this measurable gap within the quarter.
- **Profit vs Quantity Relationship** – Scatter Plot Large volumes don't guarantee profit, with losses beyond **-5K** observed. Achievable product mix adjustments ensure relevant and sustainable margins.
- **Orders by States** – Horizontal Bar Chart **Maharashtra (87)** and **Madhya Pradesh (82)** lead, while Uttar Pradesh lags. Targeted promotions can achieve balanced growth by year-end.
- **Monthly Sales Trend** – Line Chart Sales dropped from **61K to 13K**, showing seasonal decline. Time-bound corrective actions are required to stabilize measurable monthly revenue.
- **Sales Distribution by Sub-Category** – Tree map **Printers (58K)**, **Bookcases (57K)**, and **Sarees (54K)** drive revenue. Focusing on these relevant sub-categories ensures achievable ROI growth.
- **Total Amount by Sales** – Map of India **Bangalore**, **Ahmedabad**, and **Delhi** dominate sales distribution. Expanding into **Tier-2** cities is specific, measurable, and time-bound for market expansion.

8.2. DIAGNOSTIC ANALYTICS:

- **Electronics** underperformed compared to **Clothing and Furniture**, indicating weak demand or pricing issues. This gap suggests the need for promotional strategies or product repositioning.
- **High sales volumes** did not consistently translate into profits, with several loss points observed. This highlights inefficiencies in cost management or discounting practices.
- **Maharashtra and Madhya Pradesh** show strong order volumes, while states like Uttar Pradesh and Rajasthan lag. Regional disparities suggest uneven market penetration and require targeted campaigns.
- Sales declined sharply from **61K to 13K** over time, pointing to seasonal demand drops or operational bottlenecks. This trend signals the need for corrective action to stabilize revenue.
- **Printers, Bookcases, and Sarees** dominate revenue, while smaller items contribute minimally. This imbalance indicates over-reliance on a few sub-categories, risking portfolio vulnerability.
- Major cities like **Bangalore, Ahmedabad, and Delhi** drive sales, while Tier-2 cities remain underutilized. This geographic concentration suggests untapped opportunities in emerging markets.

8.3. PREDICTIVE ANALYTICS:

- **Clothing and Furniture** are projected to sustain near-target performance, while **Electronics** may continue to underperform unless corrective action is taken.
- **Profit margins** will remain inconsistent at higher volumes; predictive modelling suggests discount optimization can stabilize profitability.
- **Maharashtra and Madhya Pradesh** are expected to maintain strong demand, while **Uttar Pradesh and Rajasthan** may stay weak without targeted campaigns.
- Sales are likely to decline further if **seasonal** demand is not addressed; festive promotions could reverse the downward trend.
- **Printers, Bookcases, and Sarees** will continue as top contributors, while smaller sub-categories will remain negligible unless diversified.
- **Major cities** will dominate future sales, but Tier-2 cities show growth potential that could increase overall share by **20%** in the next year.

8.4. PRESCRIPTIVE ANALYTICS:

- Strengthen promotional campaigns for **Electronics** to close the **4K** target gap and balance category performance.
- Optimize discounting and cost structures to ensure profitability scales consistently with higher sales volumes.
- Launch targeted marketing in weaker states like **Uttar Pradesh and Rajasthan** to achieve balanced regional growth.
- Implement seasonal promotions and supply chain improvements to reverse declining monthly sales trends.
- Prioritize investment in high-value sub-categories such as Printers, Sarees, and Bookcases while diversifying product offerings.
- Expand distribution networks into Tier-2 cities to unlock untapped market potential and increase overall sales share.

9. CONCLUSION:

This analysis shows strong performance in Clothing and Furniture, while Electronics needs corrective strategies to meet targets. Regional demand is concentrated in major states and cities, with clear opportunities in Tier-2 markets. Profitability and sales trends require optimization, guiding a roadmap for balanced growth and sustainable expansion.
